INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

(School of University of Engineering and Management, Kolkata)

End Semester Examination, April – May, 2025

Degree: **BCA** , Semester : **Fourth**

Stream: **BCA**

Paper Code: **BCAAE401**

Paper Name: **Software Engineering**

Full Marks: **100** Duration: **3 Hours**

Part - A

Attempt any 10 questions out of 15 questions

Each question carries 2 Marks (2 × 10)

	1.	What is a 'work breakdown structure'?	2
	2.	What is meant by 'software prototyping'?	2
	3.	What is a software process model?	2
	4.	What is Software Engineering?	2
	5.	What is meant by 'software validation'?	2
	6.	What are the advantages of the Agile methodology?	2
	7.	Define 'technical debt' in software engineering.	2
	8.	What is the role of a project manager in software engineering?	2
	9.	Define the term 'Software Development Life Cycle' (SDLC).	2
	10.	What is 'pair programming' in Agile methodology?	2
	11.	What is the purpose of code reviews?	2
	12.	What is 'refactoring' in software development?	2
	13.	What is the significance of 'user acceptance testing'?	2
	14.	What is a 'gantt chart' in project management?	2

15. What is a software requirement specification (SRS)? 2

Part - B

Attempt any 6 questions out of 9 questions
Each question carries 5 Marks (5 × 6)

16. What is data dictionary? Is there any use of data dictionary in Data Flow Diagram (DFD)? 5

17. What do you mean by test case design? Differentiate between white box and black box testing in detail with the help of an example. 5

18. What are design patterns? What are the advantages of using design patterns? Name some popular design patterns. 5

19. What do you mean by the terms cohesion and coupling in the context of software design? How are these concepts useful in arriving at a good design of a system? 5

20. What do you understand by the term software life cycle? Why is it necessary to model software life cycle and to document it? 5

21. How does 'Project Risk' factor affect the spiral model of software development? 5

22. Distinguish between alpha, beta, and acceptance testing. How are the test cases designed for these tests? Are the test cases for the three types of tests necessarily identical? Explain your answer. 5

23. What do you understand by a modular design? How can you determine whether a given design is modular or not? 5

24. What is the difference between SRS document and Design document? What are the contents of both the documents? 5

Part - C

Attempt any 5 questions out of 8 questions
Each question carries 10 Marks (10 × 5)

25. What do you understand by automatic program analysis? Give a broad classification of different types of program analysis tools used during program development. What are the different types of information produced by each type of tool? 10

26. What is the difference between object-oriented analysis (OOA) and object-oriented design (OOD)? 10

27. What do you mean by modular design? What are the different factors that affect the modularity of a design? How can you assess the modularity of a design? What are the advantages of a modular design? **10**

28. Briefly explain the V SDLC model. Identify why for developing safety-critical software, the V SDLC model is usually considered suitable. **10**

29. What are the major advantages of first constructing a working prototype before starting to develop the actual software? What are the disadvantages of this approach? **10**

30. Which problem does the facade design pattern attempt to solve? What solutions does it offer? **10**

31. What do you understand by a layered software design? What are the advantages of a layered design? Explain your answer by using suitable examples. **10**

32. Make a checklist of various types of errors that might exist in an SRS document, so that this checklist can be used to review an SRS document. **10**

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